



Internship Project Report

Analysis and Identification of Improvement Areas for Wheel Rim Scratches and Surface Damages

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Abstract

In automotive and tractor manufacturing, the surface quality of wheel rims is a critical customer-facing quality attribute, as wheel rims form a large visible exterior surface and are among the first components noticed by the customer. Even minor scratches can adversely affect perceived product quality, leading to rejection during final inspection, increased rework effort, and potential impact on customer satisfaction. Therefore, minimizing wheel rim scratches is essential from both quality and brand image perspectives.

During a one-month industrial internship at Escorts Kubota Limited, a systematic observation-based study was carried out to analyze the occurrence of wheel rim scratches across the manufacturing and inspection process. The study covered the end-to-end process flow starting from supplier handling and Supplier Quality Assurance (SQA) inspection, followed by the Receipt Inspection (RI) department, and subsequently through tyre assembly, wheel assembly, Pre-Delivery Inspection (PDI), and QCRI final inspection. Based on shop-floor observations and interaction with quality personnel, it was observed that the majority of wheel rim scratches are generated during tyre fitment and wheel assembly stages rather than during later inspection or vehicle assembly stages.

This report presents a brief analytical assessment of the factors contributing to wheel rim scratches and highlights key areas for improvement opportunities. A limited number of conceptual improvement proposals are suggested based on these observations. All recommendations are analytical in nature and were not implemented during the internship due to its limited duration.

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I am deeply thankful to my supervisors Mr. Ankur Tyagi Sir and Mr. Chaman Singh Sir for their constant guidance, support, and encouragement throughout the internship period. Their valuable insights helped me understand the overall functioning of the assembly line, inter-departmental coordination, and the systematic working of a large-scale manufacturing plant.

This internship enabled me to gain a broader understanding of industrial practices, workflow management, and the role of quality assurance within a manufacturing organization, which significantly contributed to my technical and professional development.

I would also like to express my sincere gratitude to the Department of Mechanical Engineering, National Institute of Technology, Durgapur, for providing me with this opportunity. I am especially thankful to my Faculty Advisor, Mr. Ratnesh Kumar, and the Training and Placement Cell for their support and assistance in facilitating my internship opportunity at Escorts Kubota Limited as part of my academic curriculum.

Finally, I would like to thank all the employees and staff members at Escorts Kubota Limited who directly or indirectly supported me during the internship and made this learning experience valuable and enriching.

Chapter 1 : Introduction

1.1 Background of the Study

In the automotive and tractor manufacturing industry, the surface quality of components is an important factor influencing overall product acceptance and customer satisfaction. The wheel rim is a highly visible component, and surface defects such as scratches can adversely affect the visual quality of the vehicle and lead to rework, repainting, or rejection.

During the one-month industrial internship at Escorts Kubota Limited under the Quality Assurance department, instances of scratches on wheel rims were observed during inspection stages. Although these scratches were primarily cosmetic in nature, they contributed to increased quality-related efforts and highlighted the need for systematic analysis.

This internship project focuses on identifying and analyzing the possible causes of wheel rim scratches and proposing improvement opportunities using a structured quality approach. The study is strictly observational and analytical, with no implementation of changes carried out during the internship period.

1.2 Objectives of the Project

The main objectives of this internship project are:

- (i) To understand the overall functioning of the manufacturing and assembly line in an automotive production plant.
- (ii) To observe quality assurance practices followed during different stages of production and inspection.
- (iii) To identify the stages contributing to the occurrence of wheel rim scratches.
- (iv) To propose conceptual improvement opportunities for reducing wheel rim scratches. .

1.3 Scope of Work

The scope of this project is limited to observation-based analysis conducted during a one-month industrial internship. The study includes reviewing existing quality inspection practices and understanding process flow related to wheel assembly operations.

The project does not include physical implementation, experimental trials, or validation of proposed improvements. All findings and recommendations presented in this report

are based on observations, discussions, and analysis carried out during the internship period.

Chapter 2: Company and Department Overview



2.1 Escorts Kubota Limited – Company Profile

Escorts Kubota Limited (EKL) is a well-established Indian engineering and manufacturing company formed through the collaboration between Escorts Limited (India) and Kubota Corporation (Japan). The company is best known for its strong presence in agricultural machinery manufacturing and for maintaining high standards in production and quality.

The organization follows structured manufacturing systems with a strong emphasis on safety, quality, and customer satisfaction. Safety practices are integrated into daily operations to ensure a safe working environment, while quality standards are maintained to deliver reliable and durable products.

Escorts Kubota Limited follows a systematic and technology-driven approach to manufacturing, with defined processes, standard operating procedures, and continuous monitoring of quality parameters. The company focuses on robust engineering practices and customer-oriented quality systems to improve product reliability, reduce defects, and enhance overall customer satisfaction.

2.2 Quality Assurance Department Overview

The Quality Assurance department at Escorts Kubota Limited plays a key role in ensuring that products meet defined quality standards at various stages of manufacturing. The department works in coordination with production and assembly teams to monitor quality parameters and identify potential issues.

Quality is assured at multiple stages of the manufacturing process, starting from incoming components to in-process checks and final inspection. Each stage follows defined Work Instruction Sheets (WIS) and inspection sheets, which help in maintaining quality standards for every process. This makes the Quality Assurance department one of the most important departments in the organization, as it acts as a control point for product reliability and customer satisfaction.

Quality Assurance activities include inspection, defect analysis, and review of quality trends. The department supports continuous improvement by identifying areas where process control and handling practices can be improved to reduce defects and maintain consistent product quality.

2.3 PDI and QCRI Section Description

The Pre-Delivery Inspection (PDI) section is a very important stage where the product is thoroughly checked before final clearance. At this stage, all critical aspects of the product are verified, including component fitment, correct model and part identification, driver testing, hydraulic testing, paint quality, and overall finishing. The objective of PDI is to ensure that the product meets defined quality norms and is ready for final inspection.

After successful completion of PDI, the product is forwarded to the QCRI (Quality Control and Receiving Inspection) department. QCRI acts as the final quality checkpoint, where the product is inspected once again to verify the results of previous inspections and to identify any remaining defects. This re-verification helps in ensuring consistency and reliability in product quality.

At each inspection stage, observations and results are recorded in check sheets to maintain traceability and quality control. The QCRI department prepares daily, weekly, and monthly quality reports, which highlight defect trends, recurring issues, and areas requiring attention. These reports play a critical role in understanding quality performance and provide valuable data for improving processes and enhancing product quality in the future.

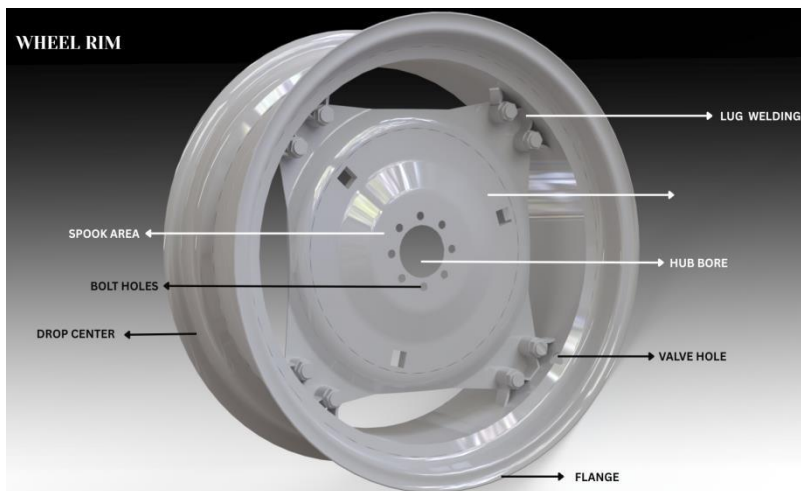
Chapter 3: Wheel Rim, Scratch & Damages Overview

3.1 Wheel Rim – Technical Definition

A wheel rim is a circular metallic component that supports the tyre and enables the wheel to be mounted onto the vehicle hub. It is designed to withstand vehicle load, maintain tyre position, and ensure proper alignment and stability during vehicle operation. Wheel rims are generally manufactured from steel and are designed with specific dimensions, bead seats, and flanges to ensure safe tyre mounting and reliable performance.

Important Parts of Wheel Rim and Their Function

- (i) The flange holds the tyre bead securely in position during operation.
- (ii) The drop center provides clearance for easy tyre mounting and removal.
- (iii) The hub bore ensures accurate centering of the wheel on the vehicle hub.
- (iv) The bolt holes allow fastening of the rim to the hub using wheel bolts.
- (v) The disc/spoke area transfers load from the rim to the hub.
- (vi) The valve hole is used for fitting the air valve for tyre inflation.



3.2 Importance of Wheel Rim Surface Finish

The surface finish of the wheel rim is an important quality parameter, as it directly affects the visual appearance and perceived quality of the product. Scratches or surface damage on the rim can lead to rejection, repainting, or rework during inspection stages.

Wheel rim surface finish is customer-critical, as the rim is a large painted and clearly visible part of the tractor. Any scratch or paint damage creates a negative first impression, which may lead to customer dissatisfaction and negative feedback. Such visible defects can adversely affect the brand image and reputation of the company.

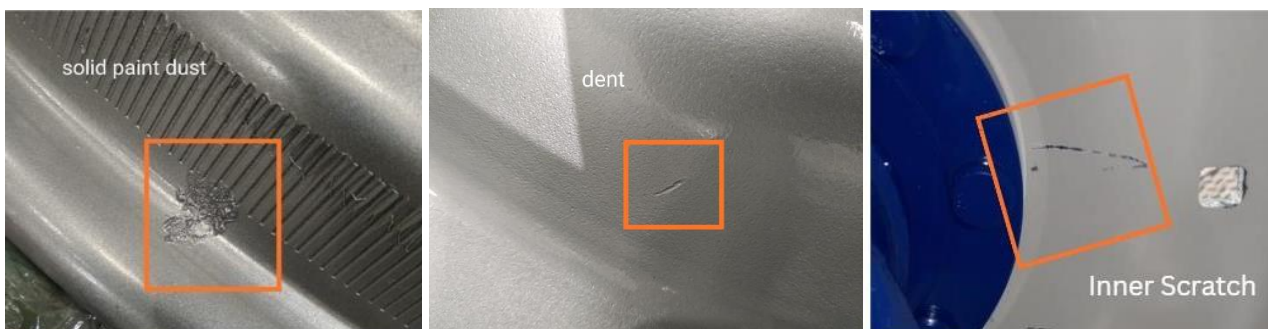
Apart from appearance, poor surface finish may also indicate improper handling or weak process control during manufacturing and assembly. Maintaining a good surface finish helps in reducing quality losses, improving customer satisfaction, and ensuring consistency in overall product quality.

3.3 Typical Scratch Locations on Wheel Rim

Based on observations during inspection, wheel rim scratches are commonly found at specific locations. These include the outer flange, bead seat area, and regions near the disc-to-rim interface. Scratches can be observed on both the outer and inner surfaces of the wheel rim, indicating exposure to damage at multiple stages of handling and assembly. Such defects are generally identified during final inspection and pre dispatch quality checks.

Since the wheel rim is a heavy component, it is more prone to scratches and damage due to multiple handling, stacking, and assembly operations. Contact with fixtures, tools, and adjacent components during these stages increases the possibility of surface damage, especially when protective measures are inadequate or misaligned. Repetitive manual handling further contributes to inconsistent contact and impact marks.

In addition to paint scratches, other wheel rim defects were also observed, such as welding spatter marks, paint dust, dents, and surface irregularities. Among these, dents were found to occur almost as frequently as paint scratches, indicating mechanical impact as a major contributor to defects. Therefore, this project focuses not only on wheel rim scratches but broadly on wheel rim surface damage and defects, with scratches being the primary defect under detailed analysis.





(v) Chapter 4: Process Flow Description

4.1 Overall Manufacturing and Inspection Process Flow

The manufacturing and inspection system at Escorts Kubota Limited follows a structured and sequential flow aimed at ensuring consistent product quality, safety, and reliability. The entire system is designed to minimize defects by integrating quality checks and documentation at multiple points, allowing early identification of deviations and maintaining traceability throughout the process.

The overall observed process flow for wheel and tyre-related components can be represented as: **Manufacturer / Supplier → Supplier Quality Assurance (SQA) → Transportation to Plant → Material Dock and Storage → Tyre Fitment → Wheel Fitment on Tractor → Pre-PDI and PDI → QCRI (Final Inspection)**

This flow reflects the coordinated interaction between manufacturing, assembly, and inspection functions. Each stage contributes to ensuring that the component condition is maintained as it moves forward, while inspection records generated at different points support quality trend analysis and continuous improvement planning.



4.2 Manufacturer / Supplier and Supplier Quality Assurance (SQA)

Wheel rims are manufactured from sheet metal at the supplier end, where they are formed into the required wheel rim shape using dedicated tools and forming operations. After forming, the rims are allowed to cool naturally before undergoing the painting process. Once painted, the rims are stacked and prepared for transportation to the assembly plant.

At Escorts Kubota Limited, wheel rims are sourced from multiple approved manufacturers, including **Bhagwati Techno Fab Pvt. Ltd.**, **Steel Strips Wheels Limited (SSWL)**, and **Carrier Wheels Pvt. Ltd.** These suppliers follow defined manufacturing and quality practices as per company requirements.

Before dispatch, the rims are inspected and verified by the Supplier Quality Assurance (SQA) team. The SQA team members are assigned by Escorts Kubota Limited and are stationed at the respective supplier manufacturing locations. Their responsibility is to ensure that the material quality, dimensions, and surface condition of the rims meet specified standards.

The role of the SQA team is critical, as inadequate inspection at this stage may result in non-conforming or damaged rims reaching the assembly line. Such issues can lead to downstream quality problems, rework, or production losses. Effective SQA control at the supplier end therefore plays a key role in preventing wheel rim scratches and damages in later stages of the process.

4.3 Transportation, Material Dock, and RI (Receiving Inspection) Department

After completion of supplier-side manufacturing and inspection, wheel rims are transported over several kilometers to Escorts Kubota Limited. Due to their large size, significant weight, and painted surface, wheel rims are vulnerable to surface damage during transportation. Although stacking and packing methods are followed to minimize movement, vibrations during transit and repeated handling can still contribute to minor scratches or surface marks before the material reaches the plant.

The material dock functions as a controlled interface between logistics and quality inspection. It is designed for unloading, segregation, and identification of incoming materials and supports smooth material flow while maintaining traceability and safety. Proper identification at this stage is essential to ensure that materials are routed correctly for inspection and further processing.

The RI (Receiving Inspection) Department is one of the most critical quality assurance departments in the plant, as it acts as the first internal quality gate after receipt of material from suppliers. The primary responsibility of RI is to verify that the received wheel rims conform to approved quality, dimensional, and surface finish requirements before they are released to downstream assembly processes.

Receiving inspection is carried out through a defined and systematic procedure rather than instant checking. When material enters the plant through the main gate, it is documented and grouped under a unique Material Receipt Number (MRN). The MRN ensures full traceability of the received lot, including supplier details, quantity, inspection status, and inspection history.

At the RI department, MRN-related inspection data is downloaded through the Oracle ERP system, which is used for material tracking and quality documentation. Based on this information, inspection responsibility is assigned to RI inspectors.

For wheel rims, inspection is performed strictly with reference to the approved Agreement of Inspection (AOI) and the corresponding engineering drawing. As observed during the internship, the sampling quantity for wheel rims is one piece per MRN, considering the size, weight, and handling difficulty of the component. The

sampled rim is inspected in detail for visual condition, dimensional accuracy, and paint quality.

The AOI specifies inspection characteristics such as visual requirements (no scratches, dents, burrs, or paint defects), dimensional parameters, inspection methods, sampling criteria, and inspection frequency. The part drawing provides detailed dimensional tolerances, sectional views, and technical notes related to the rim profile, bolt hole pattern, hub bore, and paint specifications. Both documents are simultaneously referred to during inspection to ensure conformity with approved standards.

The RI department is equipped with a range of precision measuring and inspection instruments, selected based on AOI and drawing requirements. These include vernier callipers, micrometers, height gauges, internal diameter bore gauges, slip gauges, feeler gauges, and radius gauges. Visual inspection plays a particularly important role for wheel rims due to their painted and customer-visible surface.

If the sampled wheel rim meets all specified criteria, the entire lot is approved and released for further processing. However, if any deviation or non-conformance is detected, the material is put on hold and not forwarded to the assembly line. In such cases, formal communication is sent to the supplier, and the assigned Supplier Quality Assurance (SQA) team member is involved. The SQA team coordinates with the supplier to identify the root cause of the defect and ensure corrective and preventive actions are implemented before further supplies are dispatched.

Thus, the RI department plays a vital role in preventing non-conforming wheel rims from entering the assembly process. By acting as a quality filter and a data source for supplier performance, RI significantly contributes to reducing downstream defects, rework, and quality losses, making it a key department for maintaining overall product quality.

4.4 Tyre Fitment and Wheel Fitment on Tractor

After clearance from the RI department, approved wheel rims are sent for tyre fitment. In this process, the tyre is mounted onto the rim using designated tyre fitment equipment. As the rim has a painted surface, proper alignment and controlled force are important to avoid scratches during tyre mounting. Lubrication is applied to ease tyre bead movement and ensure smooth fitment.

After tyre fitment, the tyre–wheel assembly is transferred to the wheel fitment area, where it is mounted onto the tractor hub. Correct alignment of the wheel with the hub and bolt holes is necessary to ensure proper seating. Wheel bolts are tightened in a specified sequence to maintain uniform load distribution and avoid unwanted movement during fitment. Since this stage involves multiple handling and positioning

operations, careful handling is essential to prevent surface damage and ensure quality at the final assembly stage.

4.5 Pre-PDI, PDI, and QCRI Final Inspection

After completion of assembly and wheel fitment, the tractor is directly moved to the Pre-Delivery Inspection (Pre-PDI) stage without any intermediate waiting. Pre-PDI acts as an initial quality gate where safety, electrical, and mechanical checks are carried out. During mechanical inspection, the overall condition of wheel assemblies is visually observed, and any visible wheel rim scratches, paint damage, or dents result in the tractor being held for rework. All findings are recorded in standardized checklists to maintain traceability.

Once cleared from Pre-PDI, the tractor enters the PDI stage, where functional and performance-based inspections are conducted. This includes road test verification, driver testing, hydraulic testing, and leakage inspection using ultraviolet light. These tests subject the wheel and rim to operational conditions, which may expose existing surface defects. If wheel rim scratches or damages are detected, corrective actions such as repainting or part replacement are performed before further approval.

After PDI, final verification is carried out to ensure that all parts fitted correspond to the correct tractor model as per the checklist and system records. Since wheel rims are visually prominent components, special attention is given to their surface condition to ensure compliance with appearance and quality standards before customer delivery.

The final inspection stage is QCRI (Quality Control and Receiving Inspection), which serves as the last quality checkpoint before dispatch to yards or showrooms. In QCRI, the tractor is inspected thoroughly from A to Z using detailed checklists, with wheel rim scratches and damages treated as customer-critical defects. The QCRI department maintains daily, weekly, and monthly defect trend reports, which provide essential data for analysis and form the primary basis for the present project on wheel rim scratches and damages.

Chapter 5: Problem Definition and Study Framework

5.1 Identification of the Problem

The identification of the problem was carried out using QCRI (Quality Concern Rejection Index) reports and defect trend charts related to tractor wheel and tyre fitment. From the QCRI quality trend, it was observed that scratch and surface damage defects consistently contributed to the overall QCRI, with wheel rim–related issues appearing repeatedly in defect category analysis.

Further review of Scratch & Damage DPI trends and Pareto charts showed that wheel rim scratches remained a significant and recurring defect, particularly associated with tyre fitment and wheel assembly operations. Based on this data-supported observation, wheel rim scratching during fitment was identified as a key quality problem requiring focused analysis and improvement suggestions.

5.2 Identification of VOC (Voice of Customer)

In this project, the Voice of Customer (VOC) is represented by the QCRI (Quality Control and Receiving Inspection) department, as it acts as the final internal customer before product dispatch. QCRI expectations include scratch-free wheel rims, acceptable surface finish, and compliance with defined visual quality standards.

Although multiple quality concerns are monitored by QCRI, wheel rim scratch and damage consistently remained among the top concerns and did not reduce to the required level when compared with other defects. Trend and Pareto analysis indicated that, while several issues showed improvement over time, scratch-related defects continued to contribute significantly to QCRI, highlighting it as a critical customer concern requiring focused attention.

5.3 Goal Statement

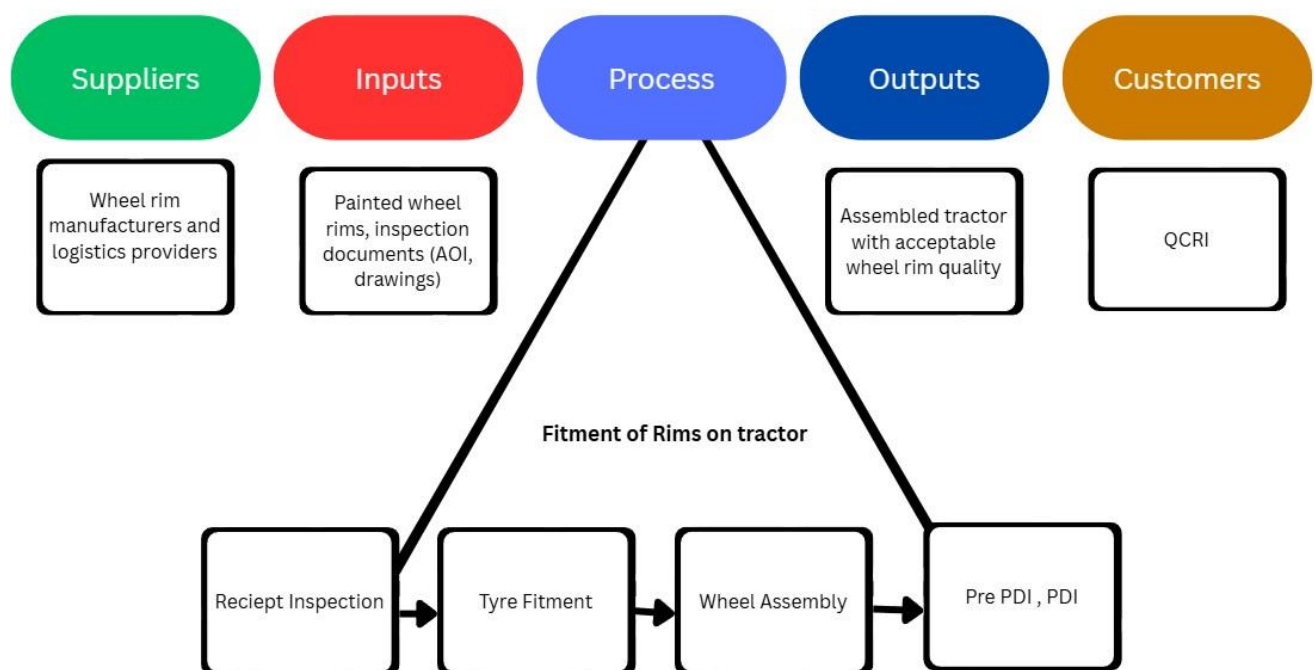
The goal of this study is to analyze the occurrence of wheel rim scratches and surface damages based on QCRI inspection data and on-floor observations, with reference to the target defect level of **0.005 DPT**. The study aims to identify the major contributing stages and potential causes responsible for these defects. Based on the analysis, conceptual improvement opportunities are proposed to support reduction of wheel rim scratch-related defects toward the defined quality target. The objective of this work is limited to analysis and recommendation, without implementation, due to the observational nature and limited duration of the internship.

5.4 Scope of the Study

The scope of this project is limited to observation and analysis of wheel rim handling, assembly, and inspection processes within the manufacturing system.

- (i) The study includes supplier-side handling methods, transportation of wheel rims, Receiving Inspection activities, tyre fitment, wheel assembly, Pre-Delivery Inspection, and QCRI inspection stages
- (ii) The scope of this work does not include implementation of improvement actions, physical trials, tooling or fixture modifications, or cost validation activities
- (iii) No additional cost, time, effort, or manpower was involved during the study, as the analysis was carried out using existing processes, standard documents, inspection records, and quality reports available within the organization.

5.5 SIPOC Diagram



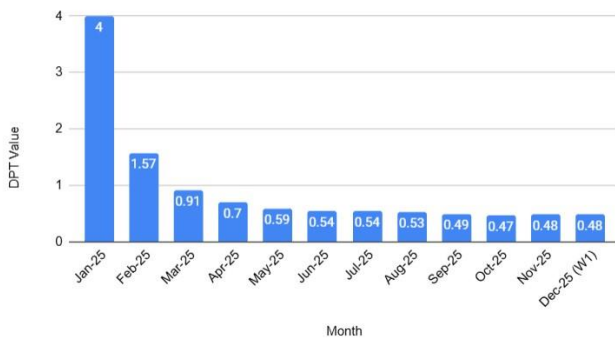
5.6 Importance of Addressing Wheel Rim Surface Defects

Wheel rim scratches and damages result in rework, repainting, inspection holds, and quality report escalations, which increase quality effort and pose reputational risk to the organization. Since wheel rims are visually prominent components, surface defects negatively affect perceived product quality and brand image. Addressing this issue through systematic observation and improvement proposals can support cost avoidance, improved inspection clearance rates, and higher customer satisfaction, making it a relevant quality improvement opportunity.

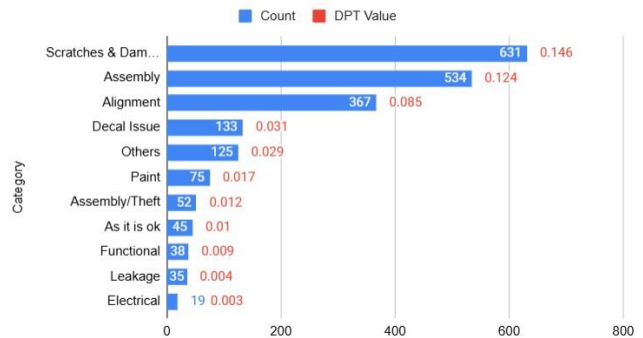
(vii) Chapter 6: Observation and Data Review

6.1 Quality Control Receiving Inspection (QCRI) Overview

FT QC RI Quality Trend (Plan - 0.05)



Category Wise DPT Nov 2025

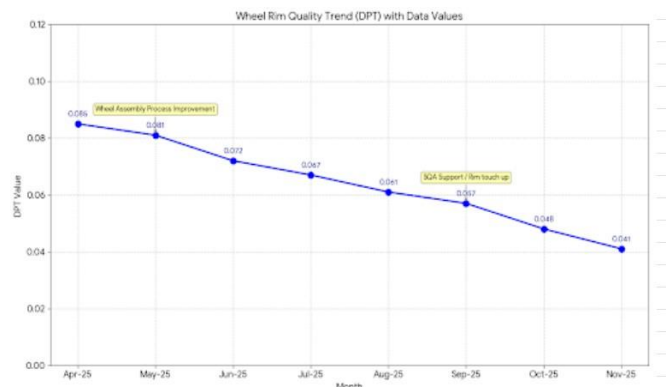
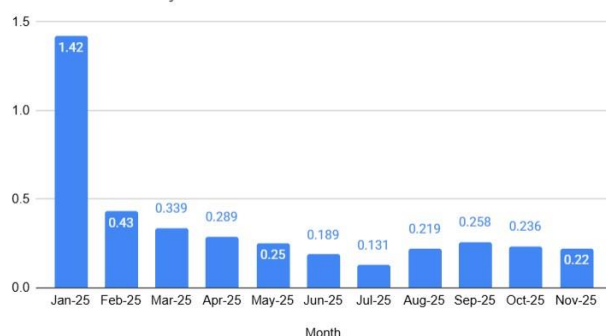


Quality Control Receiving Inspection (QCRI) plays a critical role in monitoring and controlling product quality at the receiving, assembly interface, and inspection stages. All observed defects are recorded and analyzed using Defects Per Tractor (DPT), which represents the average number of defects identified on each tractor inspected. This metric allows uniform comparison of quality performance across different months, components, and defect categories.

During the internship period, QCRI data was reviewed to understand the baseline quality condition of the manufacturing process. The collected data includes monthly DPT trends, category-wise defect distribution, and component-level Pareto analysis. Since the internship duration was limited, the analysis is focused on observed trends and statistical distributions rather than detailed experimental validation.

The QCRI system provides statistically meaningful insights by aggregating defects over a large production volume. This ensures that dominant defects are identified based on frequency and impact, making the observations reliable for proposing improvement actions.

DPT Trend Objective - 0.005



6.2 Overall Quality Trend Based on DPT

Statistical analysis of monthly DPT values shows a substantial reduction in overall defects per tractor over the observed period. In the initial months, the DPT value was significantly high, indicating a greater number of defects per unit. Over time, a sharp downward trend is observed, followed by gradual stabilization at lower DPT levels.

From the trend data, the overall DPT reduced by more than 85–90% from its peak value to the most recent months. This indicates that existing corrective actions and inspection controls have positively influenced overall quality performance. However, month-to-month variation in later stages suggests that the process is not yet fully stabilized.

Despite the improvement, the achieved DPT values remain above the defined quality objective. Statistically, this indicates that while the mean defect level has reduced, variability still exists within the process. This justifies the need for targeted, defects-specific improvement rather than general corrective actions.

6.3 Defect Distribution and Pareto Analysis



Pareto analysis of QCRI defect data highlights a highly skewed defect distribution, where a limited number of defect categories contribute to the majority of total defects. Statistical comparison shows that scratches and surface damage defects alone contribute the largest share of total DPT.

Component-level Pareto charts for Oct–Nov 2025 show that wheel rim scratches account for the highest defect count and DPT contribution, exceeding other components by a significant margin. In several months, wheel rim defects contribute more than 35–40% of total surface-related defects, clearly identifying them as the dominant issue.

This Pareto-based statistical prioritization confirms that improvement efforts focused on wheel rim handling and assembly interaction would yield the maximum reduction in overall DPT. Other defect categories, while present, have a comparatively lower statistical impact on total quality performance.

6.4 Key Observations from Data Review

Based on statistical trends and defect distribution analysis, several key observations were made. First, the reduction in overall DPT confirms that quality performance is improving; however, the persistence of certain defects indicates localized process weaknesses rather than systemic failure.

Second, the concentration of defects around wheel rims suggests a strong correlation with manual handling, orientation, and assembly contact points. The consistent appearance of rim scratches across multiple months indicates that this issue is repeatable and statistically significant, not random in nature.

Finally, since no major changes in material specifications were observed during the review period, the defect pattern statistically points toward process and method-related causes. These observations form a strong analytical basis for proposing design and process improvements in subsequent chapters.

Chapter 7: Analysis of Causes

7.1 Overview of Observed Causes and Analysis Approach

Based on the defect trends and Pareto analysis discussed in Chapter 6, wheel rim scratches were identified as the most dominant quality issue. To understand the underlying reasons for this recurring defect, a structured cause analysis was carried out. Within the given amount of time, the analysis was based on detailed process observation, interaction with operators, review of Quality Control Receiving Inspection (QCRI) records, and comparison with standard wheel assembly practices.

The cause analysis focuses on identifying process-related, material-related, and human-related factors that may contribute to rim scratches during handling and assembly. Rather than isolating a single root cause, the analysis recognizes that the defect results from multiple interacting causes occurring at different stages of the manufacturing and assembly process.

7.2 Paint Shop Handling and Inspection-Induced Scratches

During observation of the paint shop process, it was noted that wheel rims are held using a metal rod inserted from the inner side (C-side) of the rim. While this method supports the rim during painting, it creates a contact zone where paint coverage is not achieved, leading to incomplete coating in that region. This exposed area becomes more vulnerable to scratches during subsequent handling.

Additionally, before the rims are cleared for transportation, they undergo multiple inspection stages where the rims are rotated for visual checking. Continuous rotation while the rim is in contact with hard supports can result in long, continuous scratch marks along the surface. These scratches are not impact-related but are caused by sliding contact during rotation.

It was also observed that after paint touch-up operations, hand impressions are sometimes found on the rim surface. This occurs due to a lack of awareness between consecutive operators, where freshly touched-up rims are handled without adequate curing time or visual indication. This indicates a gap in communication and handling discipline at the paint shop stage.



7.3 Transportation and Stacking-Induced Scratches and Dents

Wheel rims are transported over a distance of approximately 30–40 km, during which they are stacked horizontally one above another with separators placed between adjacent rims. While separators are intended to prevent direct metal-to-metal contact, relative movement during transit can lead to successive friction between the rim and separator surfaces.

Due to road vibrations and dynamic loading, rims experience micro-sliding and intermittent impacts, which can cause both scratches and minor dents. Rims positioned at the bottom of the stack are subjected to higher compressive loads from the upper rims, making them more susceptible to deformation and dent formation.

Additional damage risk is introduced during loading and unloading operations, where impact forces may act on the rim edges. These combined factors make transportation a critical stage for surface damage generation, especially for rims already vulnerable due to partial paint coverage.



7.4 Improper Handling at Tyre Fitment Sub-Assembly



At the tyre fitment sub-assembly stage, it was observed that even rims received in good condition from the supplier and QCRI inspection develop scratches and dents due to improper handling methods. Rims are handled using a small crane, lifting two rims placed side-by-side from a stacked arrangement.

The lifting configuration results in a net unbalanced reaction force acting parallel to the sub-assembly direction. Due to this imbalance, the rim tends to shift laterally during

lifting, causing sudden rubbing against the separator or adjacent surfaces. This sliding interaction is a major contributor to surface scratches.

It was also noted that handling precision at this stage is inconsistent, and rims are not always guided carefully during lifting and placement. This increases the likelihood of accidental contact and damage, making this stage highly prone to scratch and dent formation.

7.5 Wheel Assembly at Tractor Body – Hub Contact Damage

During the final wheel assembly onto the tractor body, a crane is used to lift and position the wheel rim onto the hub. Observations indicate that during lifting, the rim often tilts toward one direction due to the lifting geometry and load distribution.

As the tilted rim is brought closer to the hub, the inner surface of the rim comes into contact with hub bolts or studs. This unintended contact leads to scratches on the inner side of the wheel, which are difficult to detect at early stages and are often identified only during later inspections.

This issue is primarily caused by lack of alignment control during crane-assisted positioning and absence of guiding mechanisms to prevent tilt. As a result, this stage contributes significantly to inner-surface scratches despite the rim being in acceptable condition prior to assembly.



Chapter 8: Proposed Improvement Opportunities

8.1 Proposal 1: Revised Paint Shop Holding Method Using Non-Visible Rim Surface

It is proposed that the paint shop holding method be redesigned so that the rim is supported from the portion where the tyre and tube are mounted. This surface remains completely covered after tyre fitment and is therefore not visible in the final assembled condition.

Although the current scratches occur on the inner (C-side) surface, they still affect perceived quality during inspection stages. Since wheel rim scratches are classified as visual defects rather than functional defects, shifting the holding contact to a nonvisible region can eliminate appearance-related rejections.

Additionally, it is proposed to ensure proper dry film thickness (DFT) during painting so that the coating can withstand minor handling interactions. This approach reduces visible scratches without affecting functional performance.

8.2 Proposal 2: Controlled Transportation Handling and Optimized Separator Hardness

Since transportation over long distances is unavoidable, major process changes may not be practical. Therefore, it is proposed to focus on controlled loading and unloading practices, ensuring that rims are handled without sudden impacts.

Another improvement opportunity is to revise the hardness of the separator material used between stacked rims. The separator should be softer than the rim material so that, in case of friction or micro-sliding during transit, the separator undergoes wear instead of the rim surface.

In combination with adequate paint film thickness, softer separators can significantly reduce scratch and dent formation during transportation while maintaining existing logistics flow.

8.3 Proposal 3: Corrected Rim Orientation at Tyre Fitment to Eliminate Reaction Forces

It is proposed to change the rim orientation during crane lifting at the tyre fitment subassembly. Current observations indicate that an unbalanced reaction force causes lateral movement of the rim, leading to rubbing against separators and adjacent surfaces.

By reorienting the rim such that the lifting force acts only in the vertical (upward) direction, lateral reaction forces can be eliminated. This ensures that no relative slipping occurs between the rim and separator during lifting.

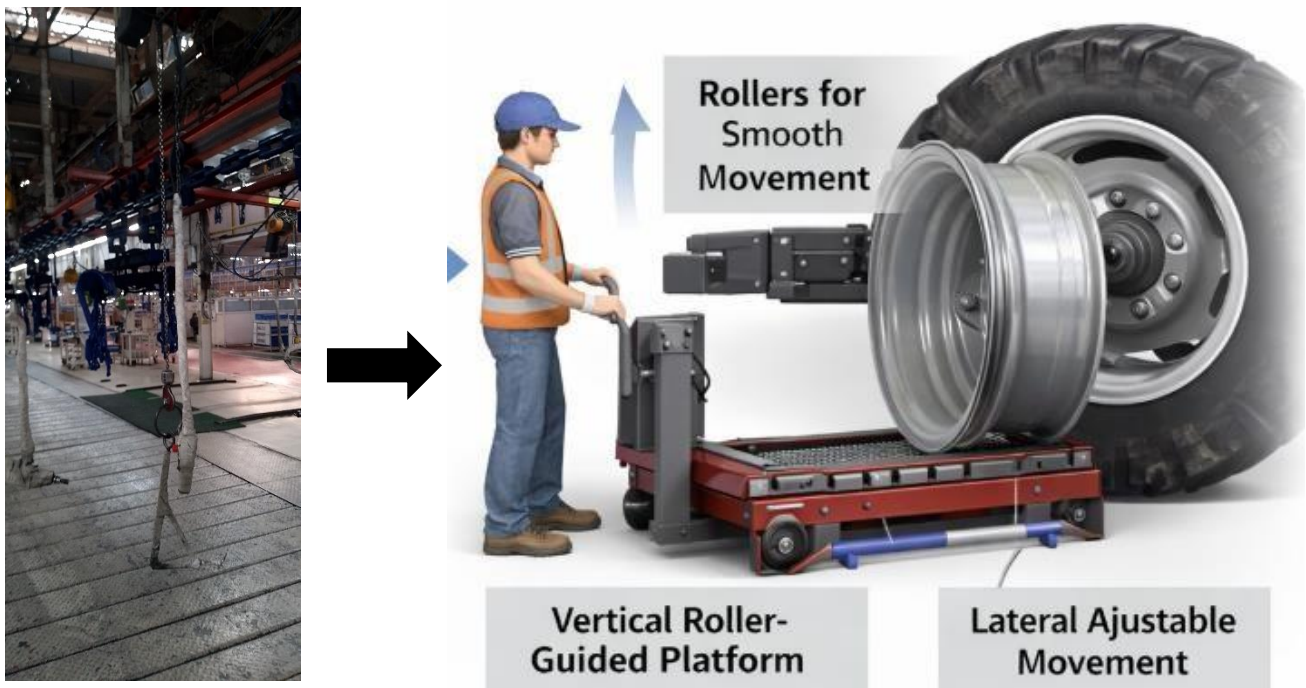
After stable lifting, the rim can then be safely transferred to the wheel assembly area, reducing scratch generation at one of the most critical defect-prone stages.

8.4 Proposal 4: Alternative Wheel Handling Method During Final Wheel Fitment

Significant internal rim damage is observed during crane-based wheel fitment at the tractor body. To address this, it is proposed to replace direct crane handling with an alternative guided handling method.

In this approach, the wheel can be rolled from storage onto a small hand-drawn lifter. The wheel is then oriented parallel to the hub face and lifted vertically using a jack mechanism. This controlled motion minimizes tilt and prevents contact between the rim inner surface and hub bolts.

Although the initial investment for such equipment may be higher, the approach offers long-term benefits through substantial reduction in rim scratches, dents, rework, and DPT, making it a high-impact quality improvement opportunity.



Chapter 9: Sustenance and Quality Assurance Considerations

Sustaining improvements in wheel rim quality requires a systematic approach that integrates process changes with ongoing monitoring and operator awareness. Among the proposed improvement opportunities, the most effective and practical starting point is the modification of rim orientation during tyre assembly. This step is particularly advantageous because it is a Kaizen-style improvement—simple, cost-free, and implementable within the existing process without requiring additional machinery or resources. By orienting the rim so that lifting forces act strictly in the vertical direction, lateral reaction forces are eliminated, preventing the rim from slipping against separators or adjacent surfaces, which is a primary cause of scratches and dents.

Beyond this immediate improvement, long-term sustenance can be achieved through a combination of procedural standardization, operator training, and consistent quality monitoring. Standard operating procedures (SOPs) should clearly define handling practices at every stage, from paint shop to final wheel fitment, ensuring that all operators follow the best practices. Regular audits and inspections can identify deviations or emerging risks, allowing timely corrective actions before defects occur. Maintaining proper paint thickness and ensuring the appropriate hardness of separator materials will further protect rims during transportation and storage.

Additionally, fostering a culture of continuous improvement among the workforce is critical. Encouraging operators to report minor handling issues or near-miss scratches can help the team refine processes over time. Integrating these measures into the quality management system will not only sustain the reduction in visual defects but also improve overall product reliability, reduce rework, and enhance customer perception of the product's quality. By combining the low-cost, high-impact Kaizen step of rim reorientation with systematic monitoring and training, the organization can achieve long-lasting improvements and ensure that the benefits of the proposed changes are fully realized.

Conclusion

The project focused on the systematic observation, analysis, and identification of improvement opportunities to reduce wheel rim scratches and surface damages across the entire tractor wheel assembly process. Wheel rim scratches, although primarily visual defects, have a significant impact on perceived product quality, inspection efficiency, and customer satisfaction. By analyzing each stage—from supplier handling to paint shop processes, transportation, tyre fitment, and final wheel assembly—it was evident that a combination of process modifications and operator awareness can lead to substantial improvements without major investments.

The study proposed four key improvement opportunities. First, revising the paint shop holding method to support the rim from a non-visible surface prevents scratches during painting without affecting functional performance. Second, controlled handling during transportation, combined with optimized separator hardness, reduces friction and micro-sliding, minimizing damage while maintaining existing logistics flow. Third, correcting rim orientation during tyre fitment eliminates lateral reaction forces during crane lifting, preventing rubbing against separators and adjacent surfaces. This Kaizen-style improvement is highly effective, cost-free, and immediately implementable, making it the most practical starting point for sustaining quality improvements. Fourth, introducing an alternative guided handling method during final wheel fitment ensures stable vertical lifting and minimizes internal rim contact, reducing scratches and rework.

Sustaining these improvements requires systematic measures, including standardizing handling procedures, providing operator training, conducting periodic audits, monitoring paint thickness and separator materials, and fostering a culture of continuous improvement. Integrating these measures into the quality assurance framework ensures that the benefits of process changes are long-lasting, defects are minimized, and overall product reliability and customer perception are enhanced.

In conclusion, careful observation, targeted analysis, and practical improvement proposals can significantly enhance wheel rim quality without incurring substantial costs. By implementing simple, high-impact process modifications—particularly the rim reorientation during tyre assembly—combined with sustained quality practices, organizations can achieve lasting reductions in visual defects, minimize rework, and maintain high standards of manufacturing excellence.

Personal Learning and Takeaways

This project has been one of the best decisions I have made, as it gave me the opportunity to bridge the gap between theoretical knowledge and the realities of a manufacturing plant. I learned how the entire system works, and how every process, part, and person is interconnected. Nothing happens in isolation—each stage depends on the previous one, and even small changes in one area can affect the overall outcome.

I also realized the importance of initiative and communication. I did not receive all the information directly; to understand a process, I had to approach different people—workers, supervisors, and line leaders—who guided me and sometimes referred me to others with better expertise. This experience taught me that asking questions is one of the most powerful tools for learning, and that knowledge in a real-world system is often distributed among many people.

Through this process, I gained hands-on knowledge about tractor parts, assembly processes, and how a large manufacturing organization functions. I made meaningful connections, understood the dependencies between different departments, and developed a deeper appreciation for teamwork and coordination. I am truly grateful for this opportunity, as it has broadened my understanding, strengthened my practical skills, and provided insights that go far beyond textbooks.